

Class 27 map bifurcations

Questions

Teams 1 and 2: Post screenshots of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

1. The last two days of our course will be progress presentations. On May 7th from 2pm-5pm (in Pierce 301) we will have final presentations.

Goals of the progress presentation:

- Introduce your topic in a way that is accessible to your classmates
- Connect your project to the course's mathematical content
- Use slides to illustrate and clarify key ideas (not as scripts or text-heavy notes)
- Clearly present your project goals, plans, and current progress

Goals of the final presentation:

- remind class of project goal
- efficiently provide necessary background
- present the core content of the modeling approach, replication work, and extension
- share results and conclusions
- provide recommendations for next steps

Examples of presentation slides

- (a) What does their project seem to be about?
- (b) Identify techniques that this team used to illustrate their ideas.
- (c) How much text did they use on their slides? What did they use it for?
- (d) If you were to suggest revising a slide, which one would you revise?
- (e) Identify a slide you thought was particularly well done.

2. (9.4.2) The tent map is a simple analytical model that has some properties in common with the Lorenz map. Let

$$x_{n+1} = \begin{cases} 2x_n, & 0 \leq x_n \leq \frac{1}{2} \\ 2 - 2x_n, & \frac{1}{2} \leq x_n \leq 1. \end{cases}$$

- (a) Look for a period-3 or period-4 point. If you find one, are such orbits stable or unstable?
- (b) If you want, you can think about whether there is a period- k orbit...
- (c) Show that, for the tent map, every periodic orbit of period $k \geq 1$ is unstable.

Use the chain rule to reason about $\frac{dg}{dx}$ for $g(x) = f^k(x)$.

The tent map has many periodic orbits, and none of them are attracting.

3. Consider the map $f(x) = r - x^2$.

(a) Draw this map for $r = 1$.

Compare it to the tent map. In what ways is this map also a unimodal stretch-and-fold map? What is different about the slopes?

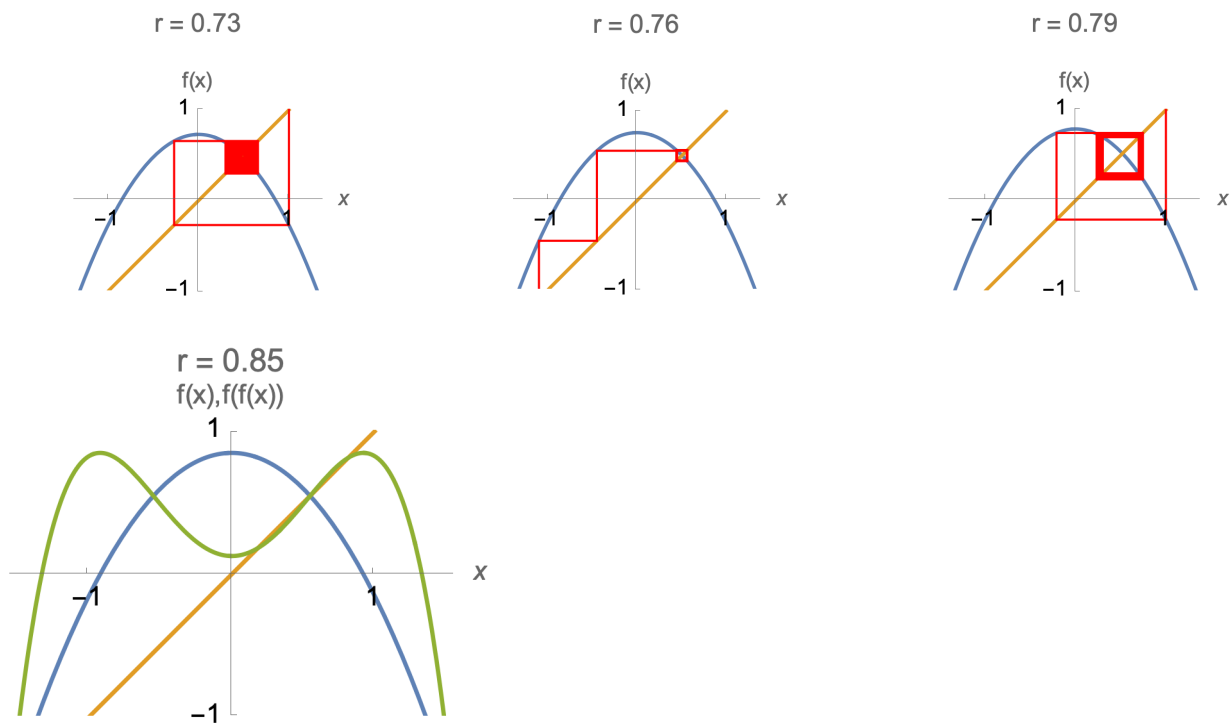
(b) Draw the map for $r = 0$. The fixed points are at $x^* = -1, 0$. What are their stabilities?

(c) As r increases, the slope becomes steeper at each fixed point. When $r = 3/4$, the right fixed point is at $x^* = 1/2$. What happens to its stability at that moment?

The stability of a fixed point changes as r increases, so we are encountering a bifurcation.

4. In the previous question, a fixed point changes stability, so it undergoes a bifurcation.

Below are plots of the map $f(x) = r - x^2$ for parameter values near the bifurcation, together with cobwebbing. There is also a graph of $f(x)$ with $f(f(x))$ a little away from the bifurcation.



(a) Before the bifurcation ($r = 0.73$), what appears to happen to trajectories near the right fixed point? (The left fixed point is not shown.)

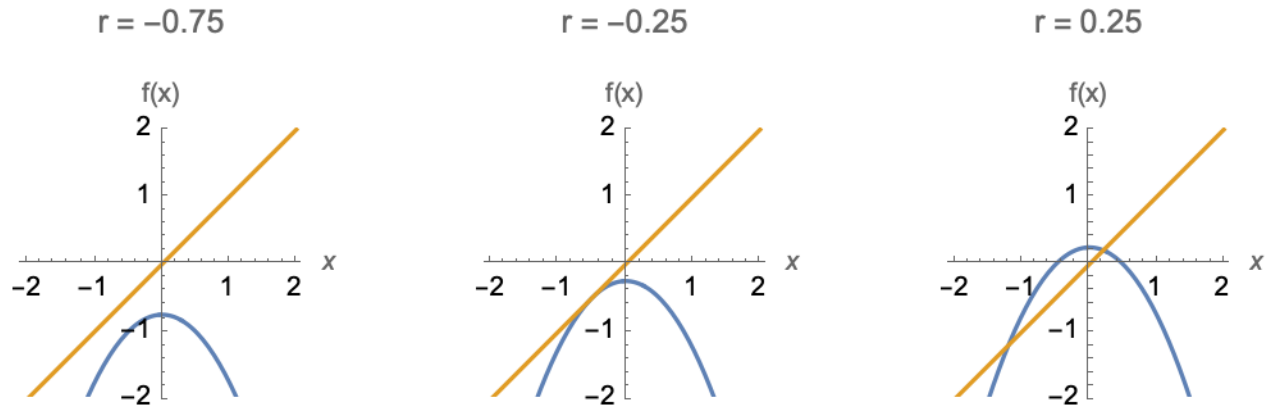
(b) After the bifurcation ($r = 0.76$ and $r = 0.79$), what appears to happen?

(c) After the bifurcation, the trajectory approaches alternation between two values. What kind of orbit does this suggest?

(d) In the graph of $f(x)$ and $f(f(x))$, locate the points involved in the period-2 orbit.

(e) The bifurcation creates a period-2 orbit as a fixed point loses stability. It is called a *flip bifurcation* or *period-doubling bifurcation*. Explain why each name makes sense.

5. Another type of map bifurcation is shown below.



- What happens in this bifurcation? What do you observe about the slopes/stabilities?
- The bifurcation is called a **saddle-node bifurcation**, a **tangent bifurcation**, or a **fold bifurcation** in the context of maps. For each of those names, how does it make sense for this bifurcation?
- At the moment of bifurcation, what is $f'(x^*)$?